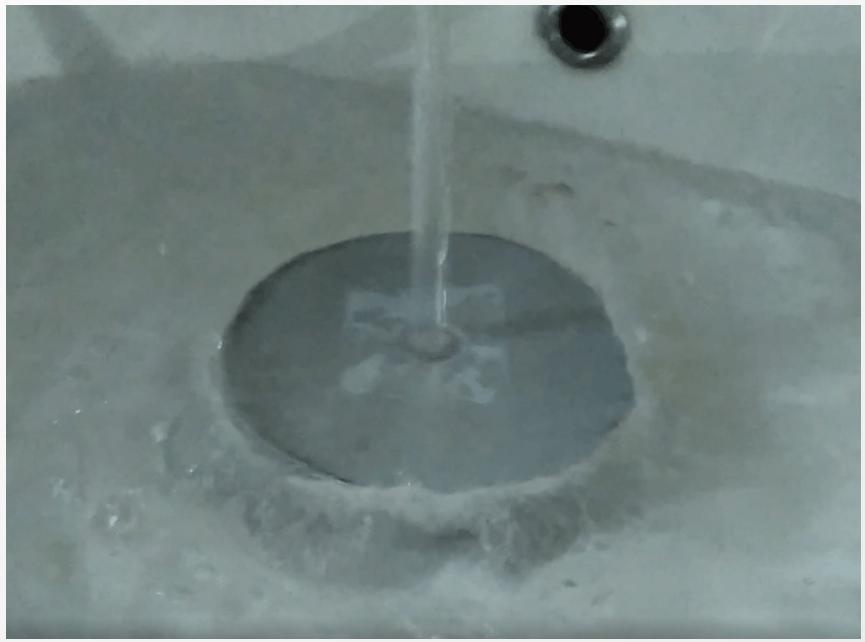
IYPT 2022

Unsinkable Disk

A metal disk with a hole in the center sinks in a water-filled container. When a strong stream of water is directed at the center of the disk, it may begin to float on the water. Explain the phenomenon and investigate the relevant parameters.



Reporter: Dachi Tchotashvili



Problem Statement



- A metal disk with a hole in the center sinks in a water-filled container. When a strong stream of water is directed at the center of the disk, it may begin to float on the water. Explain the phenomenon and investigate the relevant parameters.



Outline



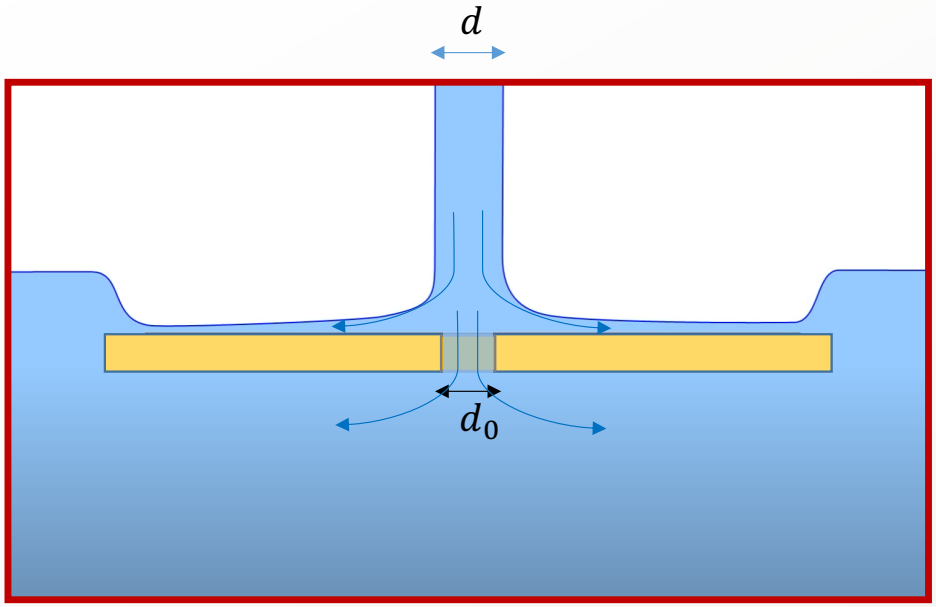
- Phenomenon explanation
- Theoretical model
- Experimental Setup
- Data taking mechanisms
- Experimental and theoretical data analysis
- Conclusion



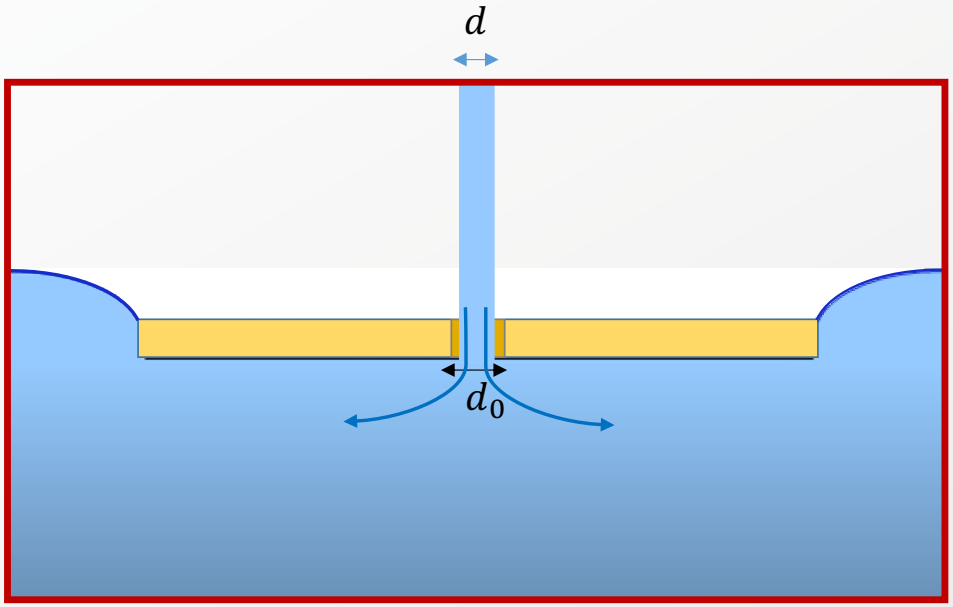


Phenomenon

Phenomenon explanation

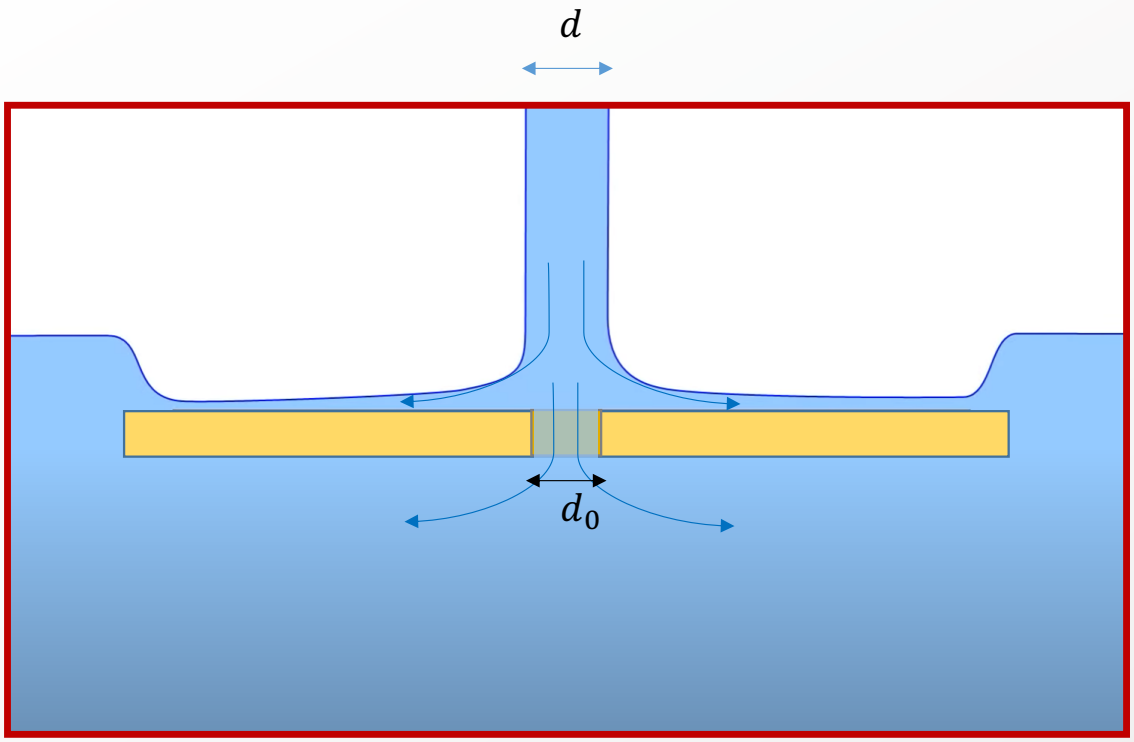


1. $d > d_0$



2. $d < d_0$

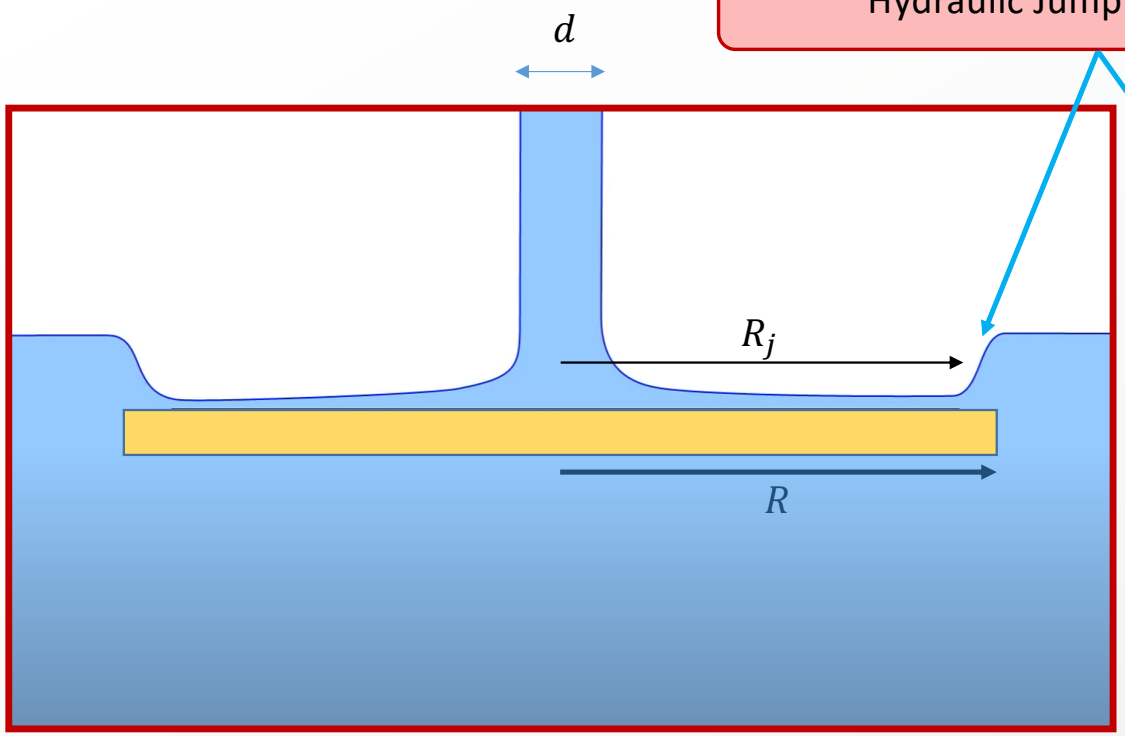
Phenomenon explanation



1. $d > d_0$

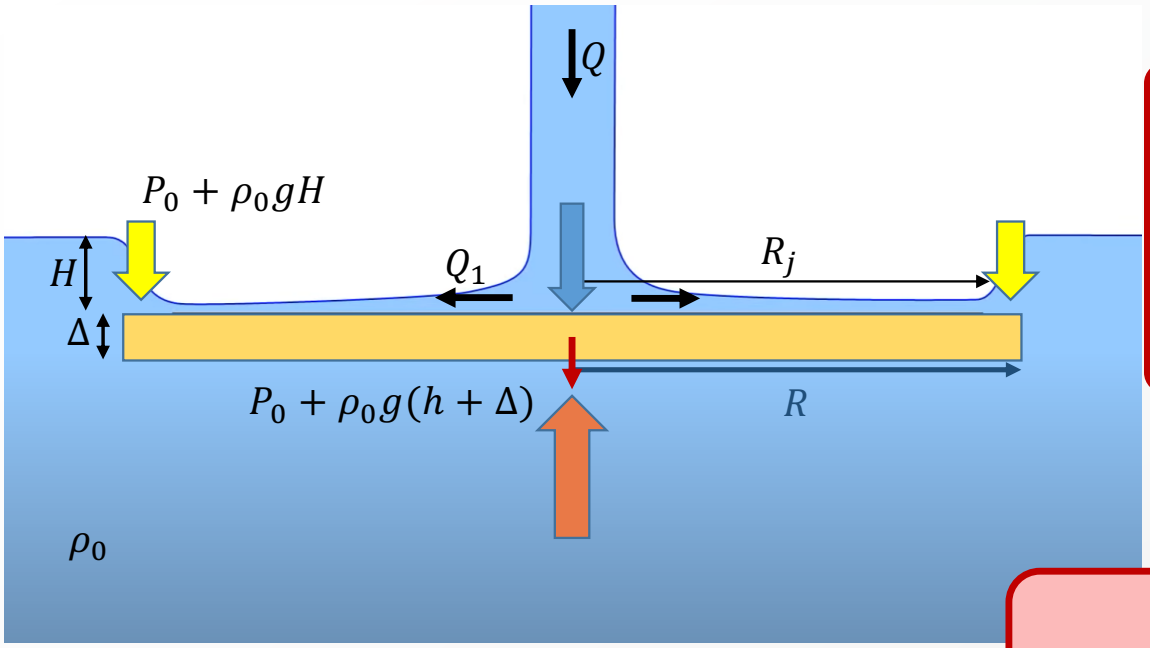
$1.\ d > d_0$

Hydraulic Jump





1. $d > d_0$ - equilibrium

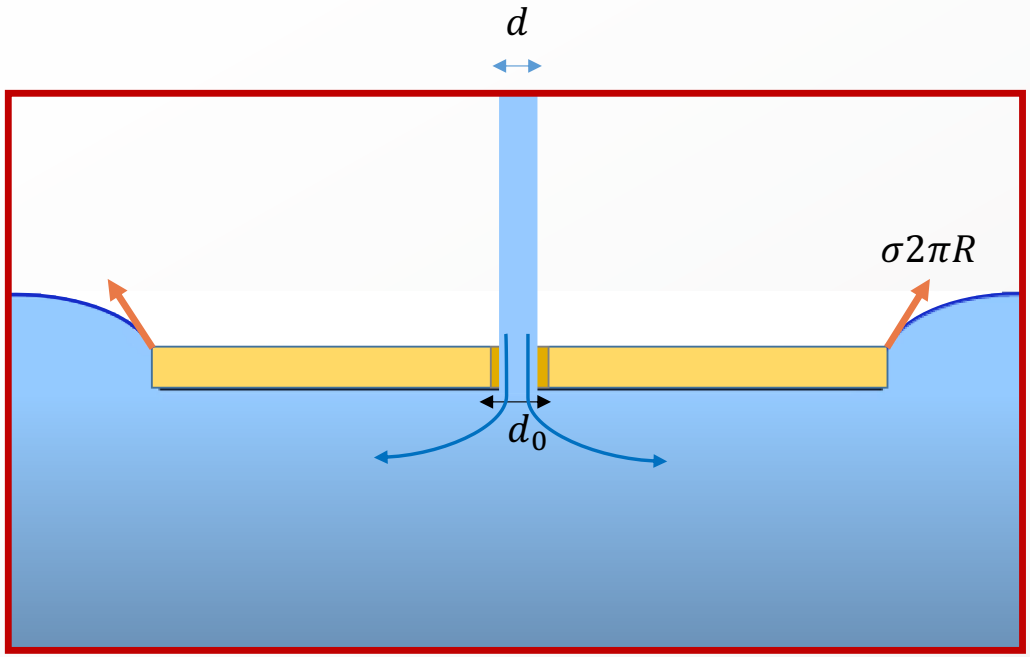


წონასწორობის პირობა:

$$\rho_0 g(H + \Delta)\,\pi R^2 = mg + \rho_0 g H \pi (R^2 - R_j^2) + F_{impact}$$

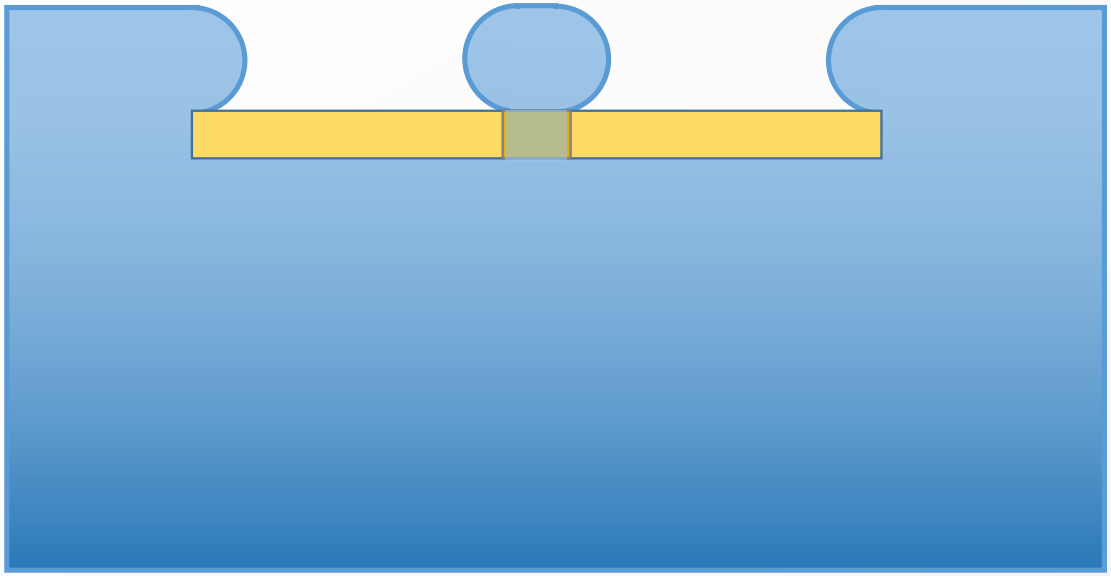
$Q\left(\frac{m^3}{s}\right)$-flow volume per second

$2.\ d < d_0$



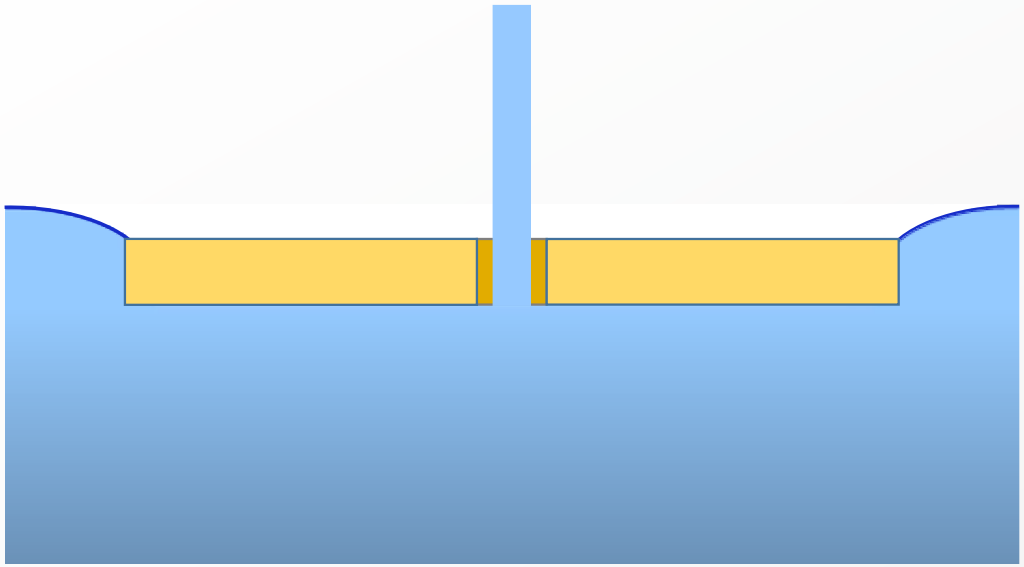
Surface Tension

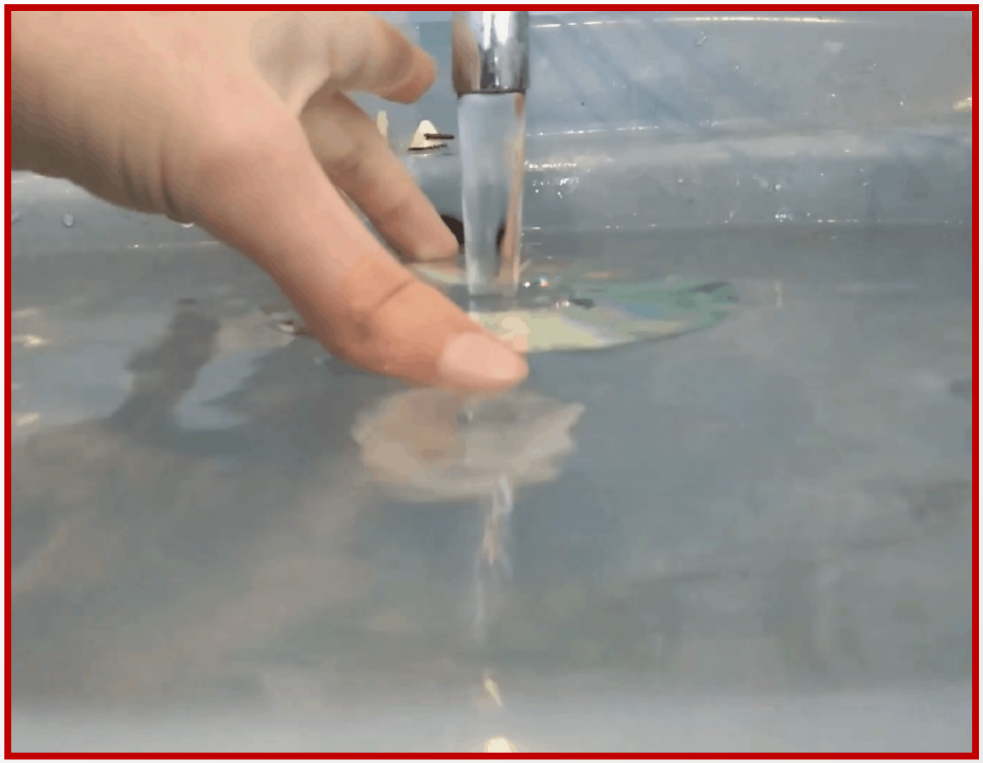
$2.\ d < d_0$



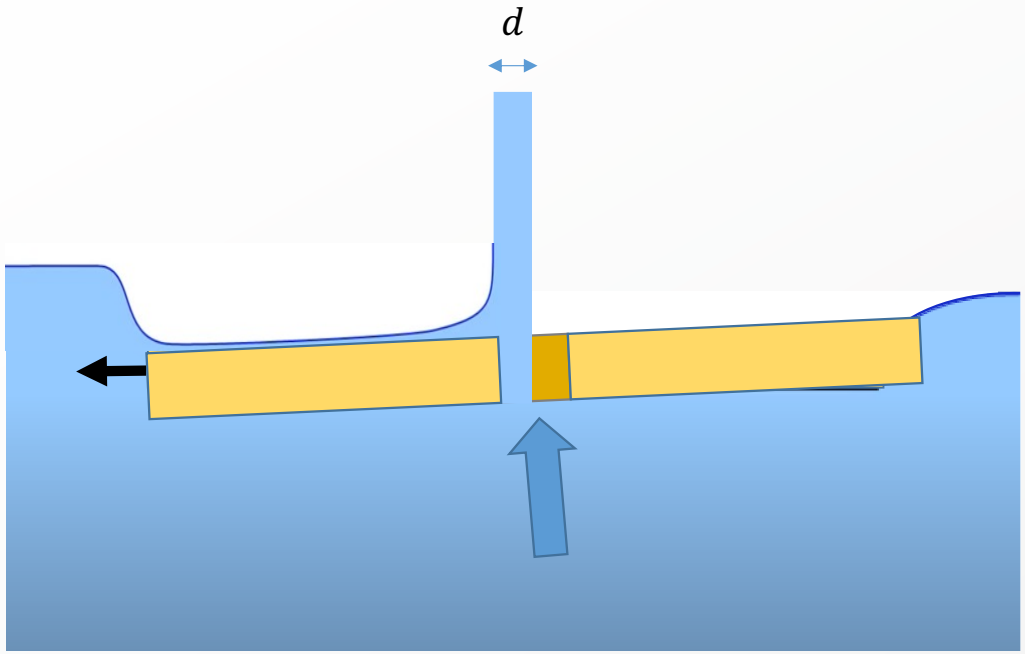


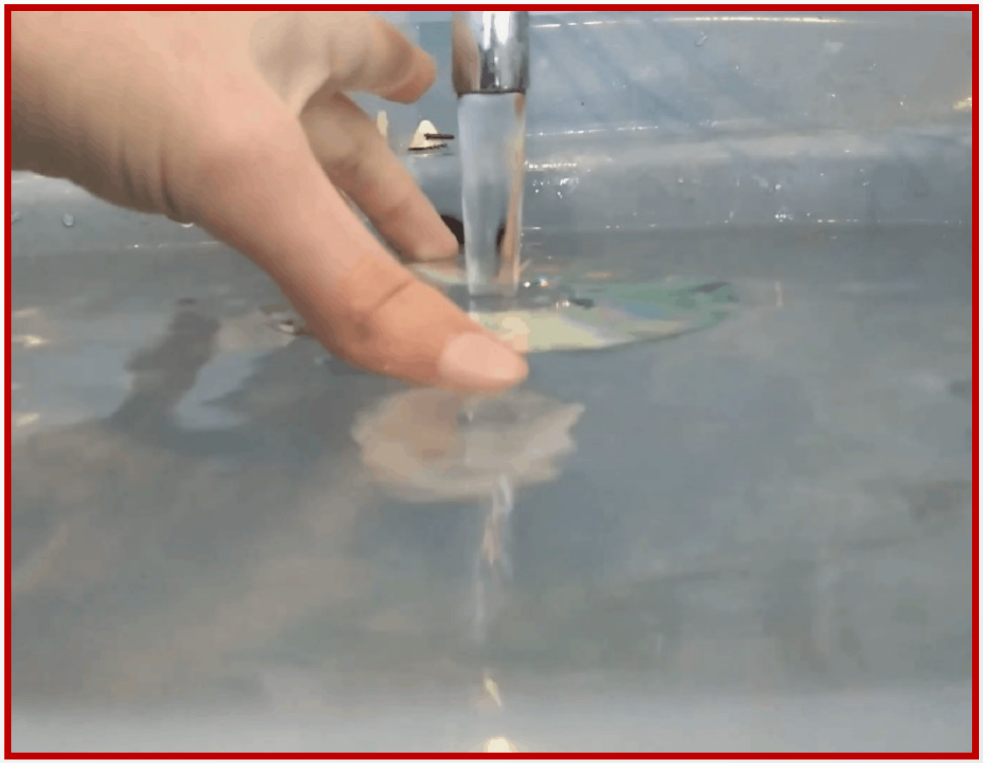
Stable equilibrium



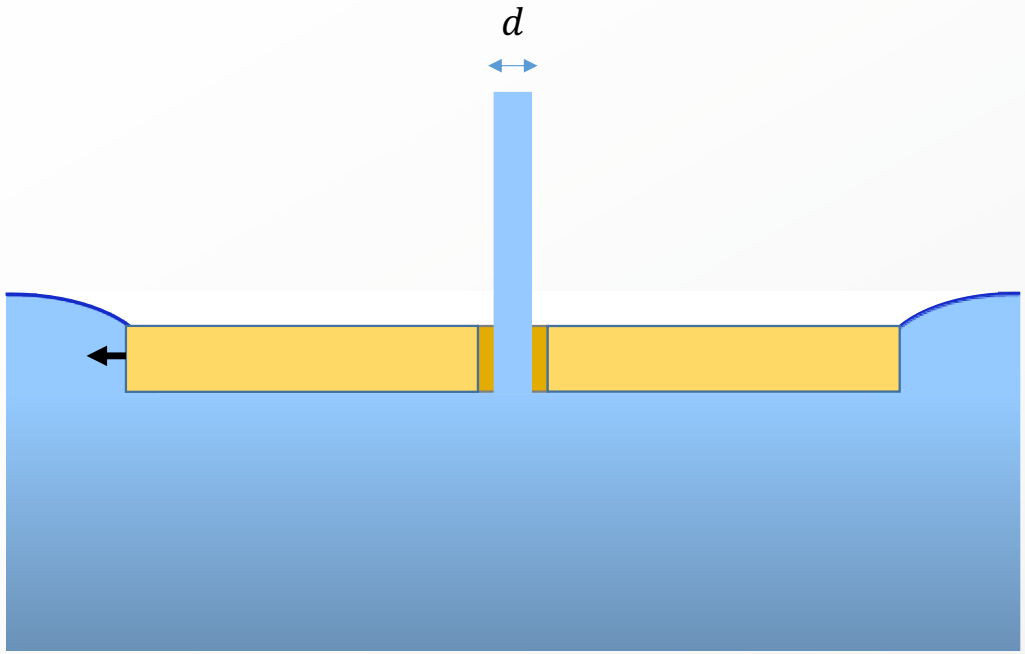


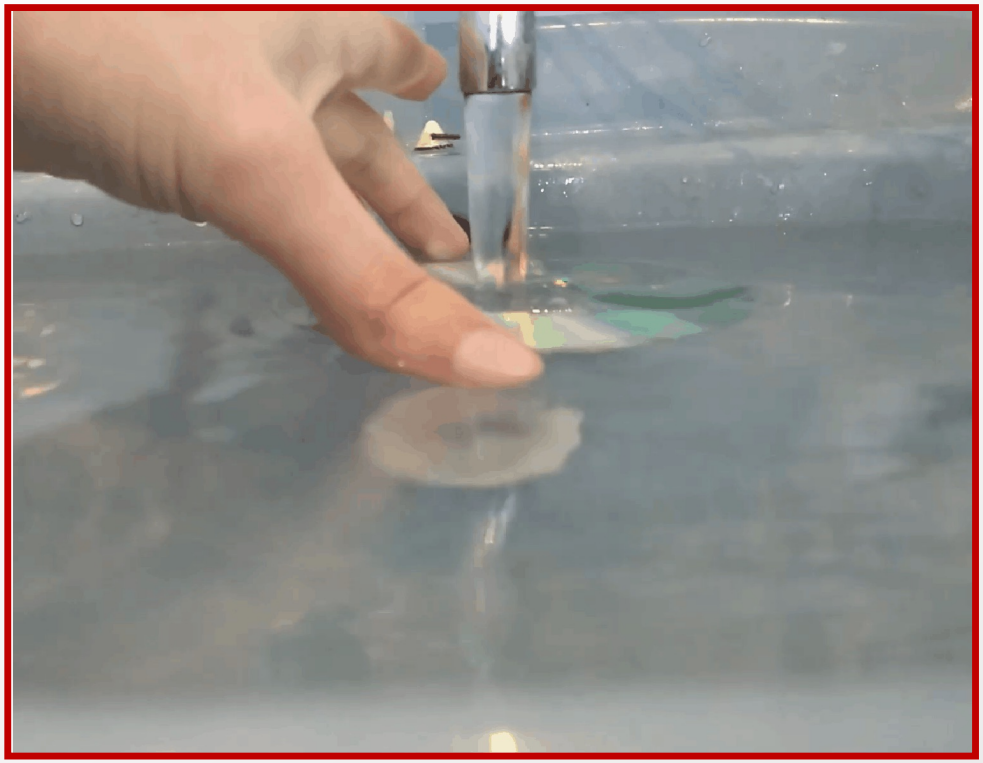
Stable equilibrium





Stable equilibrium





Stable equilibrium



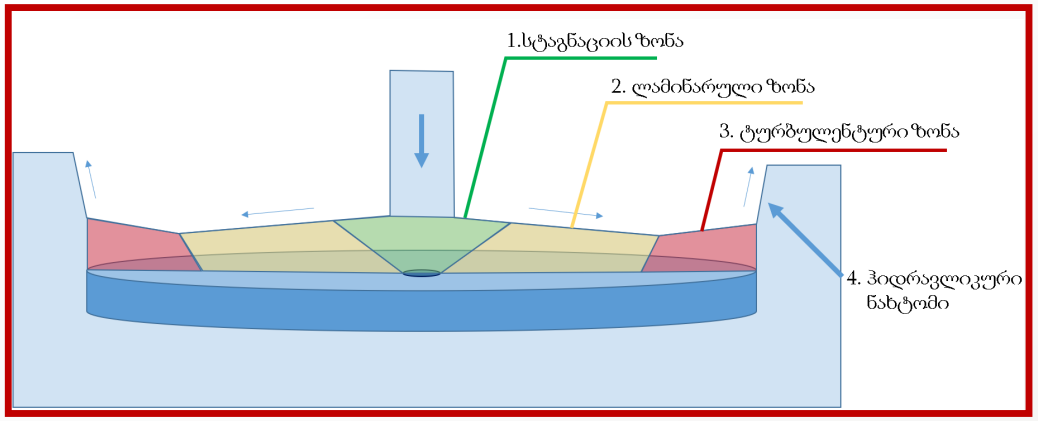




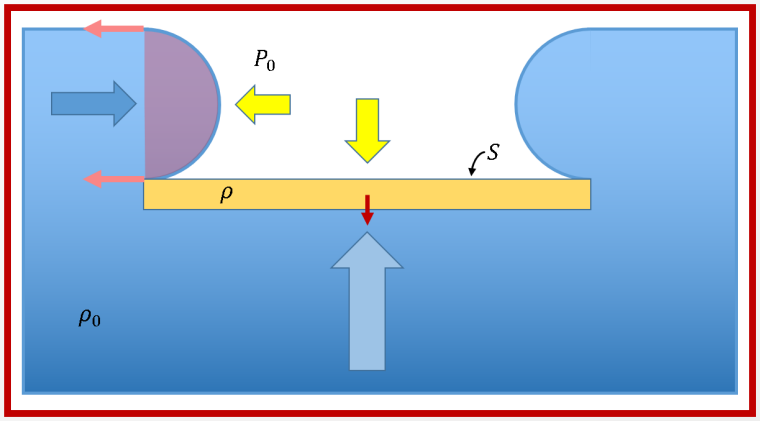
Theoretical model

Theoretical model

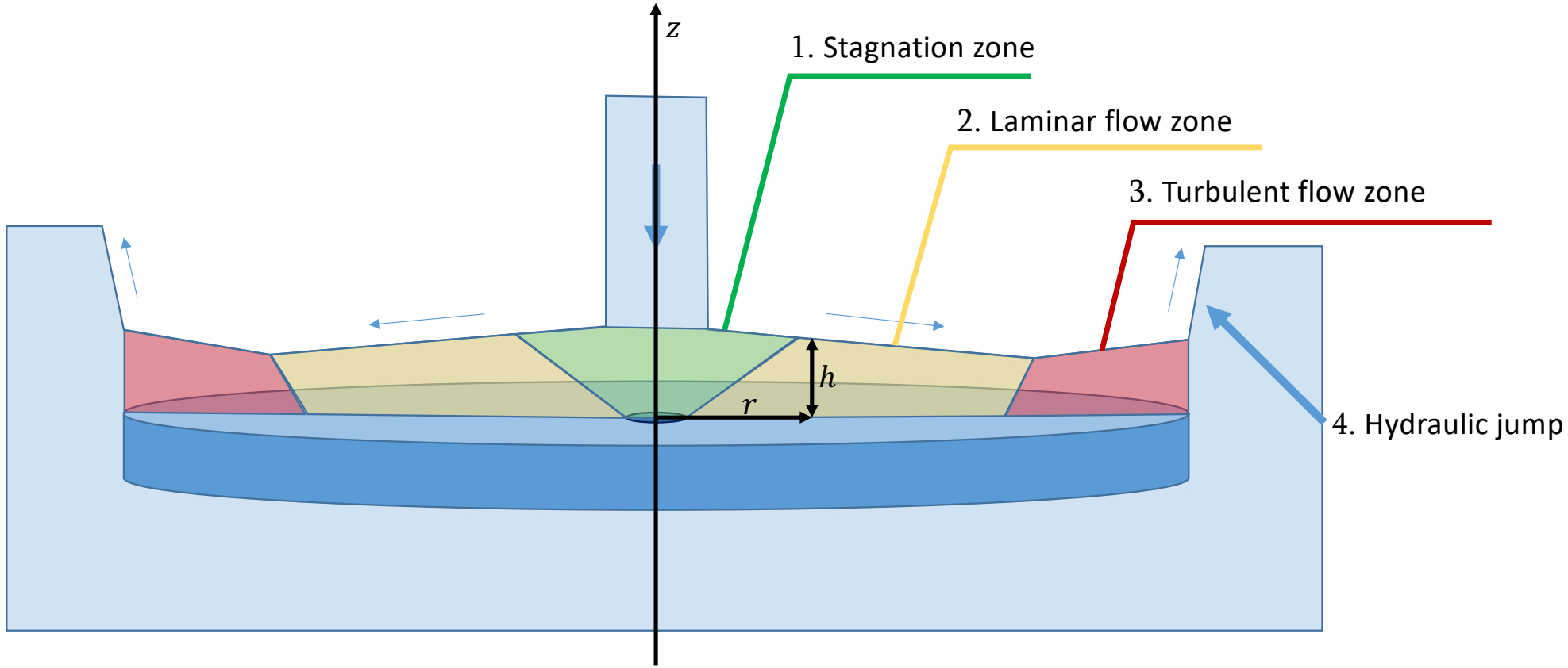
1. Flow on the surface of the disk (d>d0)



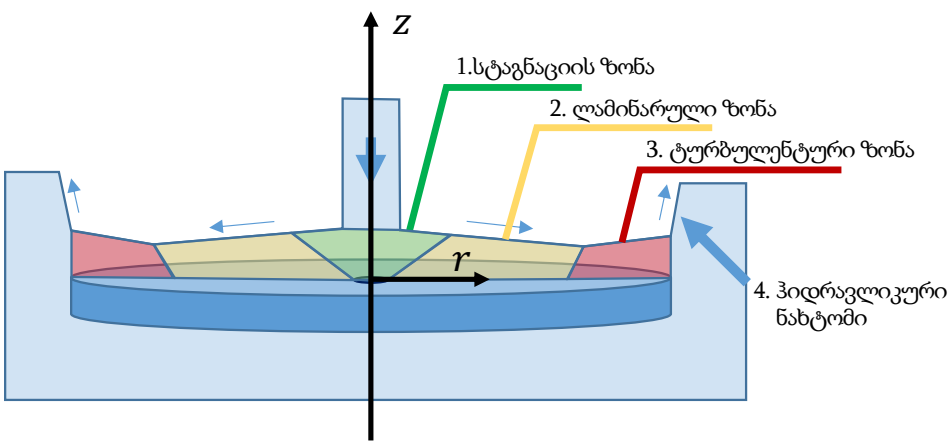
2. Equilibrium in case of d<d0



წყლის მოძრაობა დისკის ზედაპირზე



Flow on the surface of the disk



z, r *cylindrical coordinates*

$u(r,z) -$ *radial flow velocity*

$v(r,z) -$ *vertical flow velocity*

$h(r) -$ *flow thickness*

Hydrodynamic equations

$$\frac{1}{r}\frac{\delta(ur)}{\delta r} + \frac{\delta v}{\delta z} = 0 \qquad \text{continuation}$$

$$Q_1 = \int_0^h u\, dz\, 2\pi r \qquad \text{Volume conservation}$$

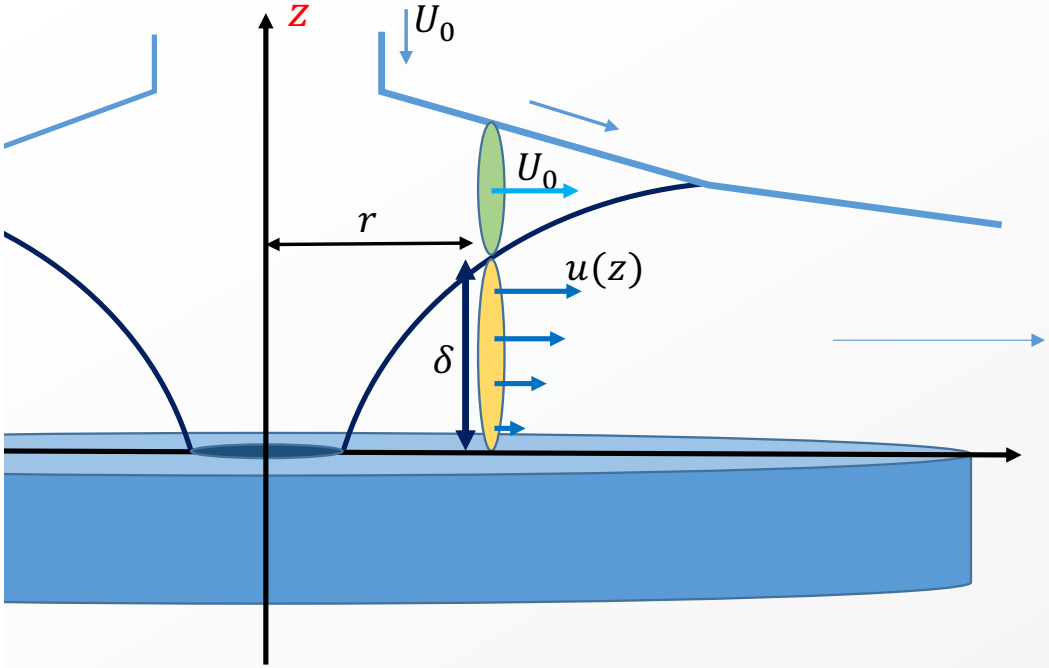
$$u\frac{\delta u}{\delta r} + v\frac{\delta u}{\delta z} = -\frac{1}{\rho_0}\frac{\delta P}{\delta r} + \frac{\mu}{\rho_0}\frac{\delta^2 u}{\delta z^2} \qquad \text{Navier stokes}$$

Boundery conditions

$u = v = 0, \quad when\ z = 0$

$\frac{\delta u}{\delta z} = 0, \quad when\ z = h$

1. Stagnation zone



Plugging in Navier-Stokes

$$\frac{1}{r}\frac{d}{dr}\int_0^{\delta} u^2 r\, dz + \frac{1}{r}\frac{d}{dr}\int_{\delta}^{h} U_0^2 r\, dz = -\frac{\mu}{\rho_0}\frac{\delta u}{\delta z}\bigg|_0 \qquad \frac{\delta P}{\delta r} = 0$$

$$u(z) = U_0 f(\eta) \qquad \eta = \frac{z}{\delta}$$

Boundary conditions

$$f(0) = 0 \qquad \int_0^1 f(\eta) = \frac{1}{2}$$

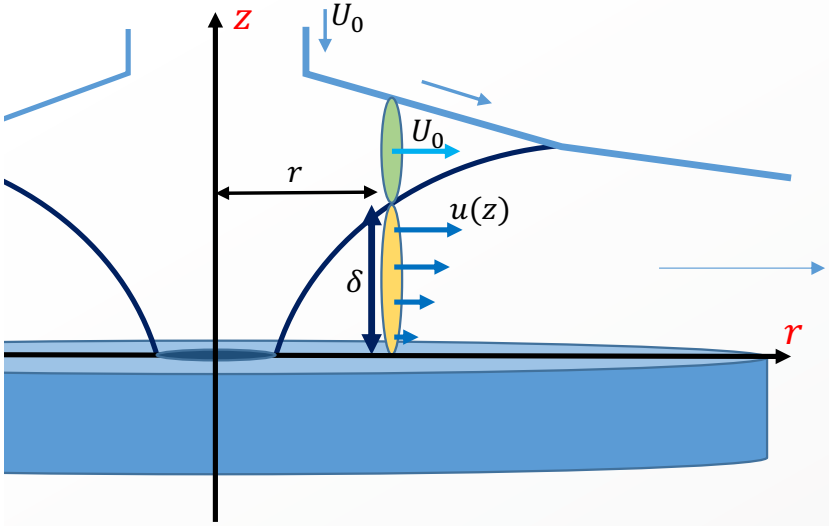
$$f'(1) = 0$$

$$f(1) = 1 \qquad f'(0) = 1$$

$$\Rightarrow \quad f(\eta) = \eta - \frac{3}{2}\eta^2 + 4\eta^3 - \frac{5}{2}\eta^4$$

Stevens, J., Webb, B.W., 1993. Measurements of flow structure in the radial layer of impinging free-surface liquid jets. International Journal of Heat and Mass Transfer 36, 3751-3758.

1. Stagnation zone



Plugging in Navier-Stokes

$$\frac{1}{r}\frac{d}{dr}\int_0^{\delta} u^2 r\, dz + \frac{1}{r}\frac{d}{dr}\int_{\delta}^{h} U_0^2 r\, dz = -\frac{\mu}{\rho_0}\frac{\delta u}{\delta z}\Big|_0 \qquad \frac{\delta P}{\delta r} = 0$$

$$u(z) = U_0 f(\eta)$$

$$\eta = \frac{z}{\delta}$$

$$\frac{1}{r}\frac{d}{dr}(U_0^2 r\delta)\int_0^1 f^2(\eta)d\eta + \frac{1}{r}\frac{d}{dr}\left(U_0^2 r(h-\delta)\right) = -\frac{\mu}{\rho_0}\frac{U_0}{\delta}f(\eta)'\Big|_0$$

$$\int_0^1 f^2(\eta)d\eta = C_1 \qquad f(\eta)'\Big|_0 = C_2$$

1. Stagnation zone

$$\frac{1}{r}\frac{d}{dr}(U_0^2 r\delta)\int_0^1 f^2(\eta)d\eta + \frac{1}{r}\frac{d}{dr}\left(U_0^2 r(h-\delta)\right) = -\frac{\mu}{\rho_0}\frac{U_0}{\delta}f(\eta)'\Big|_0$$

$$+$$

$$Q = r\int_0^{2\pi}\int_0^{\delta} u\,dz\,d\theta + r\int_0^{2\pi}\int_{\delta}^{h} U\,dz\,d\theta$$

Continuation eq.

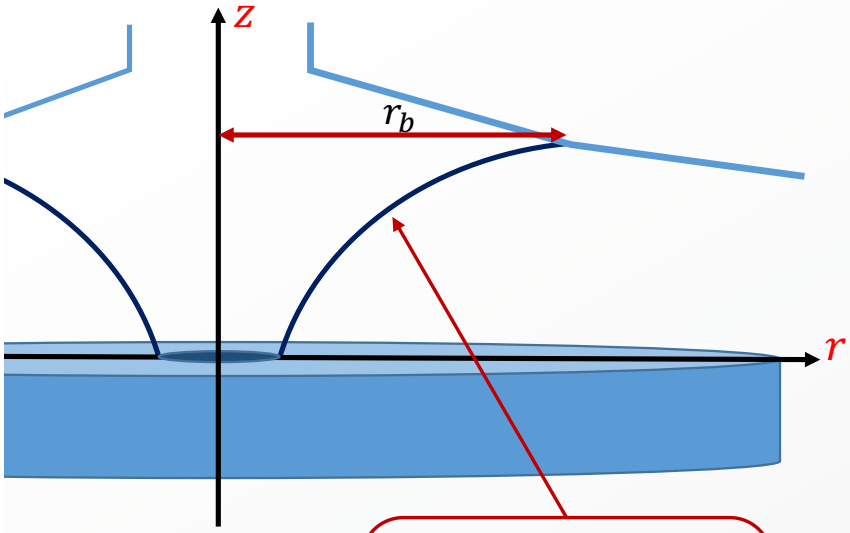
$$\delta = 2.12\sqrt{\frac{\mu\left(r-\frac{d_0}{2}\right)}{\rho U_0}}$$

$$\frac{Q}{2\pi} = rU_0\delta\int_0^1 f(\eta)d\eta + rU_0(h-\delta)$$

$$\int_0^1 f(\eta)d\eta = C_3$$

$$h = \frac{Q}{2\pi r U_0} + \delta(1-C_3)$$

1. Stagnation zone



$$\delta = 2.12\sqrt{\frac{\mu\left(r-\frac{d_0}{2}\right)}{\rho U_0}}$$

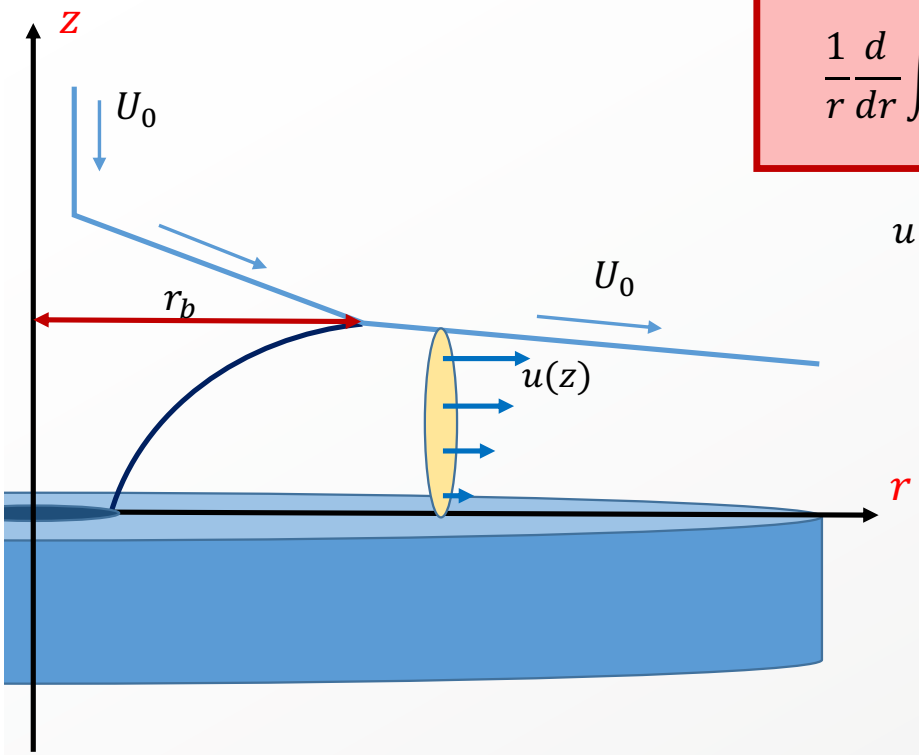
results

$$h(r) = 0.125\frac{d^2-d_0^2}{r-\frac{d_0}{2}} + \frac{1.06}{\sqrt{Re_j}}\sqrt{r-\frac{d_0}{2}}$$

$$r_b = \frac{d_0}{2} + 0.24\sqrt{d^2-d_0^2}\,Re_j^{1/3}, \text{როცა } \delta = h$$

$$Re_j = \frac{\rho U_0}{\mu}(d-d_0)$$

2. Transition to Laminar flow zone



$$\frac{1}{r}\frac{d}{dr}\int_0^h u^2 r\, dz = -\frac{\mu}{\rho_0}\frac{\delta u}{\delta z}\bigg|_0 \quad + \quad 2\pi r h U_{avg} = Q$$

$$u = U_0 f(\eta) \qquad \eta = \frac{z}{h}$$

$$r = r_b \qquad U_{avg} = \frac{U_0}{2}$$

$$h(r) = \frac{3.792}{Re_j}\frac{r^2}{\sqrt{d^2 - d_0^2}} + 0.198\,\frac{d^2 - d_0^2}{r}$$

$$U_{avg}(r) = \frac{U_0}{\frac{30.336}{Re_j}\left(\frac{r}{\sqrt{d^2 - d_0^2}}\right)^3 + 1.56}$$



Transition from laminar to turbulent zone



We postulate that the change in surface wave behaviour can be promoted by the existence of a singularity in the variation of energy with film thickness.

Dou, H.-S., Khoo, B.C., 2010. Criteria of turbulent transition in parallel flows. Modern Physics Letters B 24, 1437-1440.

$$\frac{dE}{dh} - singularity$$

Energy flow in laminar flow: $E = \rho U_{avg}^3 h = \rho \left(\frac{d^2 U_0}{8rh}\right)^3 h$

$$\frac{dE}{dh} = \left(\frac{\delta E}{\delta r}\right)\frac{dr}{dh} + \frac{\delta E}{\delta h} \Rightarrow \frac{dE}{dh} = -\frac{3(d-d_0)^6 U_0^3}{256 r^4 h^2}\left(\frac{dr}{dh}\right) - \frac{(d-d_0)^6 U_0^3}{128 r^3 h^3} \Rightarrow \frac{dE}{dh} - is\ singularity, when\ \frac{dh}{dr} = 0$$

$$r_t = 0.2964 \sqrt{d^2 - d_0^2}\, Re_j^{1/3}$$

3. Turbulent flow zone

$$\frac{u}{U_0} = \eta^{1/7}$$

$$\frac{\tau_w}{\rho_0 U_0^2} = 0.0225 \left(\frac{\mu}{\rho_0 U_0 h}\right)^{1/4}$$

(Azuma and Hoshin, 1984b; Liu et al., 1991; Schlichting, 1968).

$$\frac{dU_{avg}}{dr} = -0.04706 \left(\frac{2\pi}{Q}\right) \frac{r^{\frac{5}{4}}}{(Re_j d)^{0.25}} U_{avg}^2$$

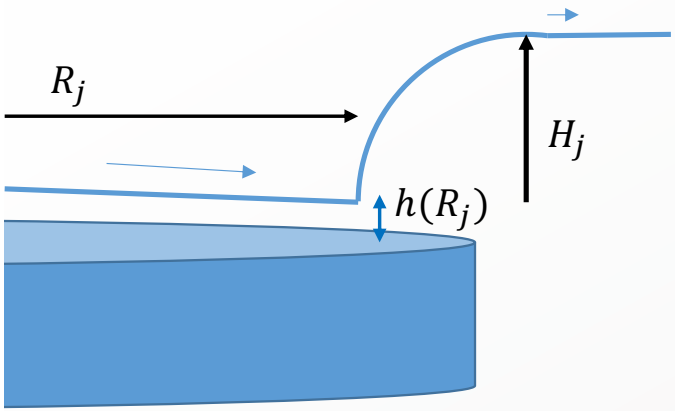
$$\tau_w = \mu \left.\frac{\delta u}{\delta z}\right|_0 - friction\ preassure\ (pa)$$

$$r = r_t, \qquad U_{avg}(laminar) = U_{avg}(turbulent) = U_t$$

$$\frac{1}{r}\frac{d}{dr}\int_0^h u^2 r\, dz = -\frac{\mu}{\rho_0}\left.\frac{\delta u}{\delta z}\right|_0 \quad + \quad 2\pi r h U_{avg} = Q$$

$$U_{avg}(r) = \frac{U_0}{\frac{0.167}{Re_j^{0.25}} \left(\frac{r}{\sqrt{d^2 - d_0^2}}\right)^{9/4} + \left(2.37 - 0.0108 Re_j^{1/2}\right)}$$

4. Hydraulic jump



$M(r) - momentum\ flow\ per\ length$

$$M_{R_j} + Ph = M_{H_j} + PH_j$$

$$H_j \gg h(R_j)$$

$$U_{avg}(r < R_j) \gg U_{avg}(r > R_j)$$

Laplace formula: $P = \sigma\left(\frac{1}{R_j} + \frac{1}{H_j}\right)$

$$M = \int_0^h \rho_0 u^2\, dz = \sigma\left(1 + \frac{H_j}{R_j}\right)$$

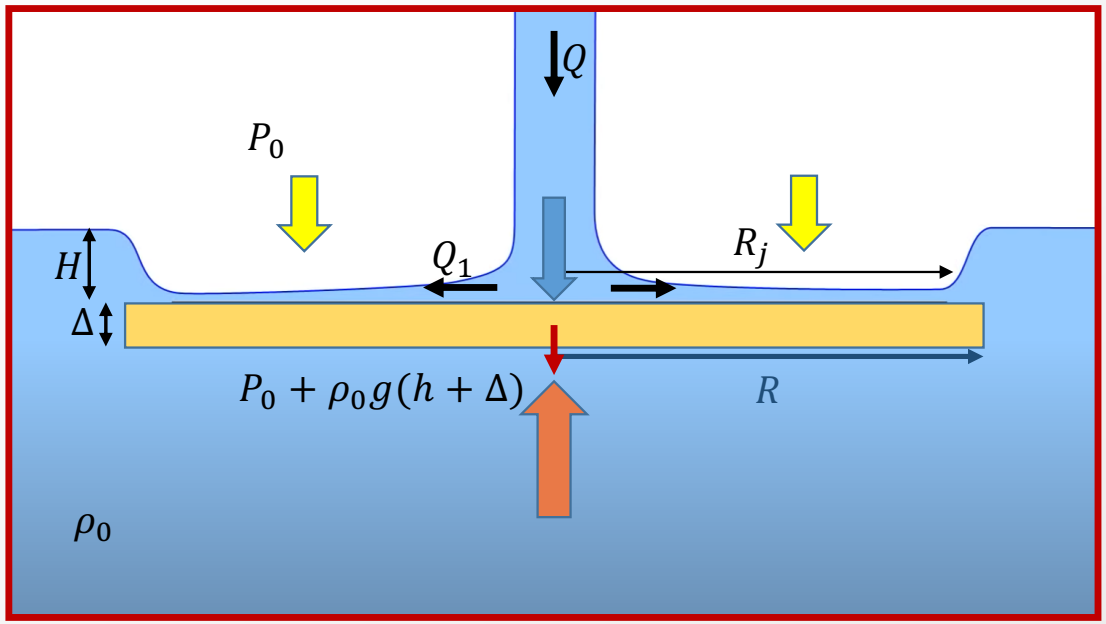
Bush, J.W., Aristoff, J.M., 2003. The influence of surface tension on the circular hydraulic jump. Journal of Fluid Mechanics 489, 229-238.

4. Hydraulic jump

$$M = \int_0^h \rho_0 u^2 \, dz = \sigma\left(1 + \frac{H_j}{R_j}\right)$$

$$\rho_0 g(H + \Delta)\,\pi R^2 = mg + \rho_0 g H \pi (R^2 - R_j^2) + \frac{4\rho_0 Q_1^2}{\pi(d^2 - d_0^2)}$$

Equilibrium condition: $H_j \geq H$



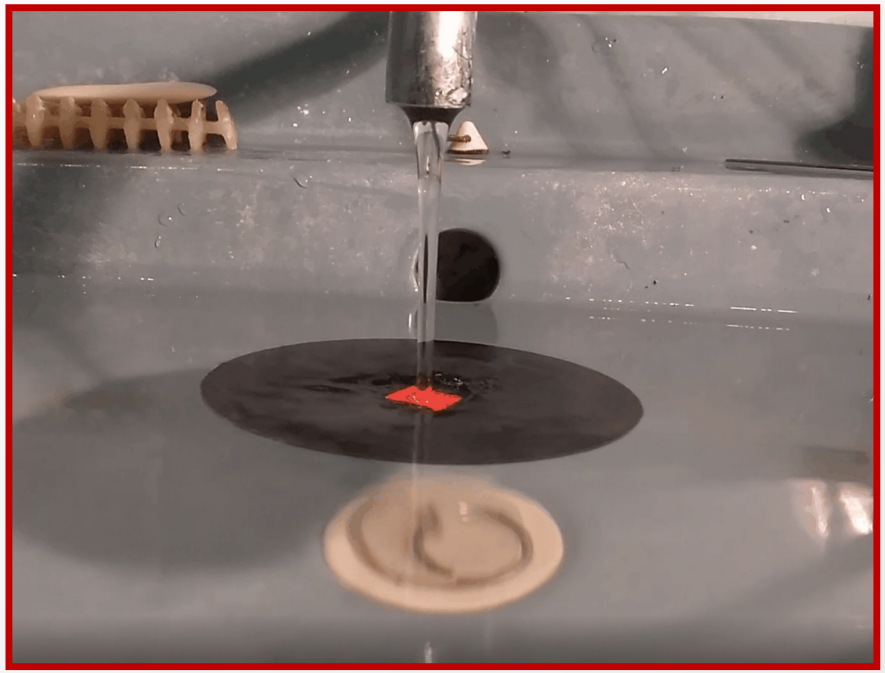
4. Hydraulic jump

$R_j < R, \qquad H_j \approx H_{j_0} \ll R$

$$M = \int_0^h \rho_0 u^2 \, dz = \sigma$$

$$H = \left(\frac{\rho}{\rho_0} - 1\right)\frac{R^2}{R_j^2}\Delta + \frac{4Q_1^2}{\pi^2(d^2 - d_0^2)gR_j^2}$$

$H_{j_0} \geq H$



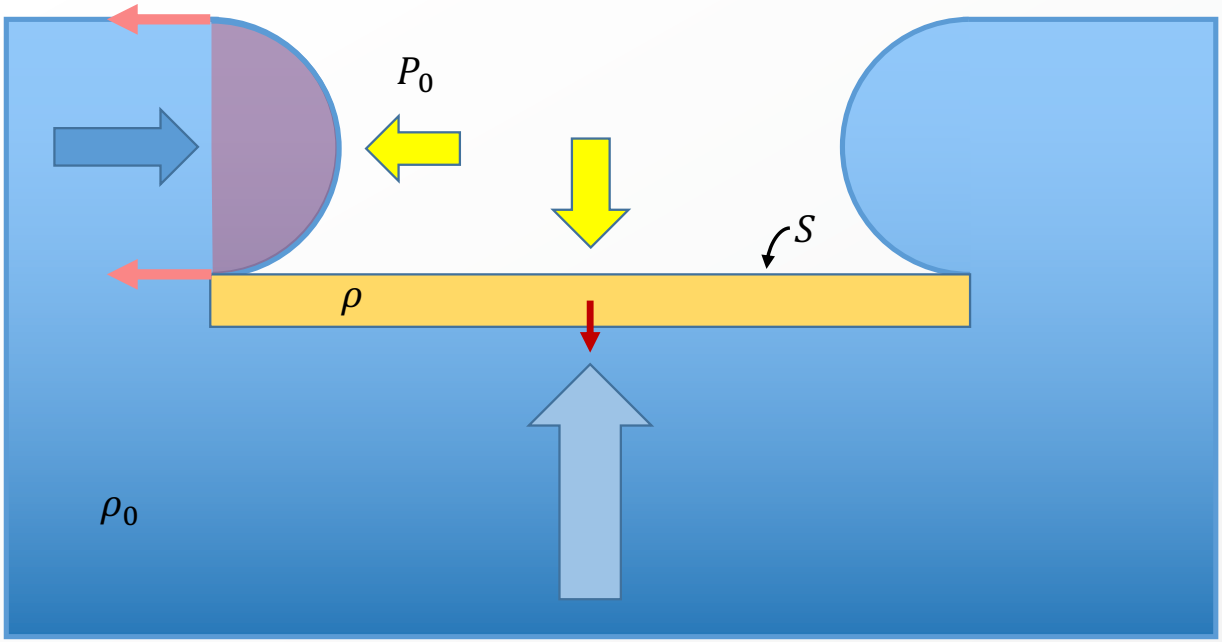
$R_j = R$

$$M = \int_0^h \rho_0 u^2 \, dz = \sigma\left(1 + \frac{H_j}{R}\right)$$

$$H = \left(\frac{\rho}{\rho_0} - 1\right)\Delta + \frac{4Q_1^2}{\pi^2(d^2 - d_0^2)gR^2}$$

$H_j \geq H$

$2.\ d < d_0$



$$\left(\frac{1}{2}\rho_0 g H + P_0\right) 2\pi R H = P_0 2\pi R H + 2(\sigma 2\pi R)$$

$$H = \sqrt{\frac{4\sigma}{\rho_0 g}}$$

$$[P_0 + \rho_0 g(H + \Delta)]S = p_0 S + \rho g \Delta S$$

$$\rho \leq \rho_0 \left(\frac{1}{\Delta}\sqrt{\frac{4\sigma}{\rho_0 g}} + 1\right)$$





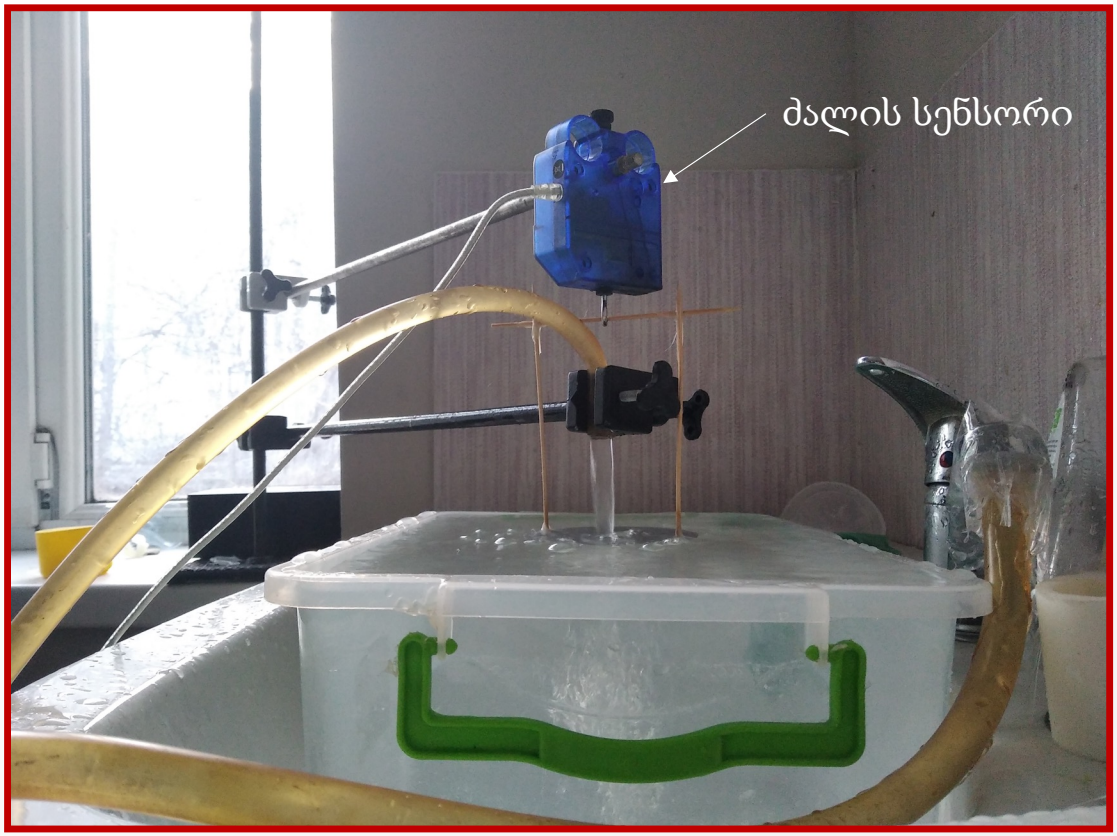
Experimental setup

Setup

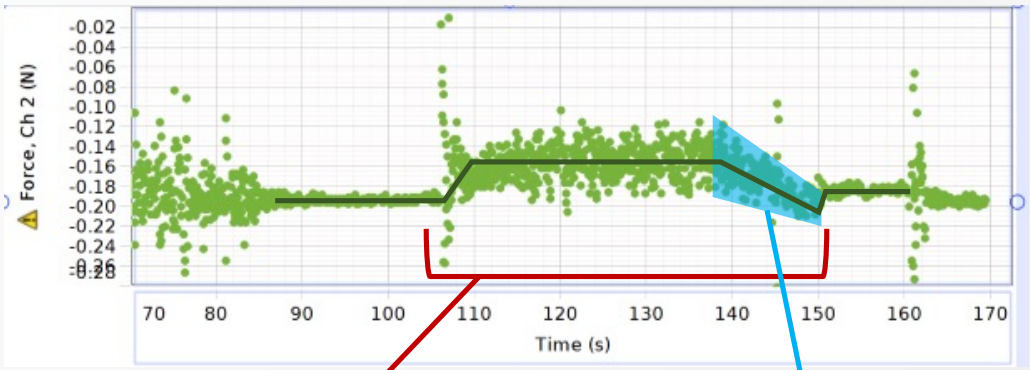


- Varied flow rate: Q
- Used vessels with different depths

Calculating force acting on the disk



$$Q = 4\frac{litre}{min}\ (maximum\ flow)$$

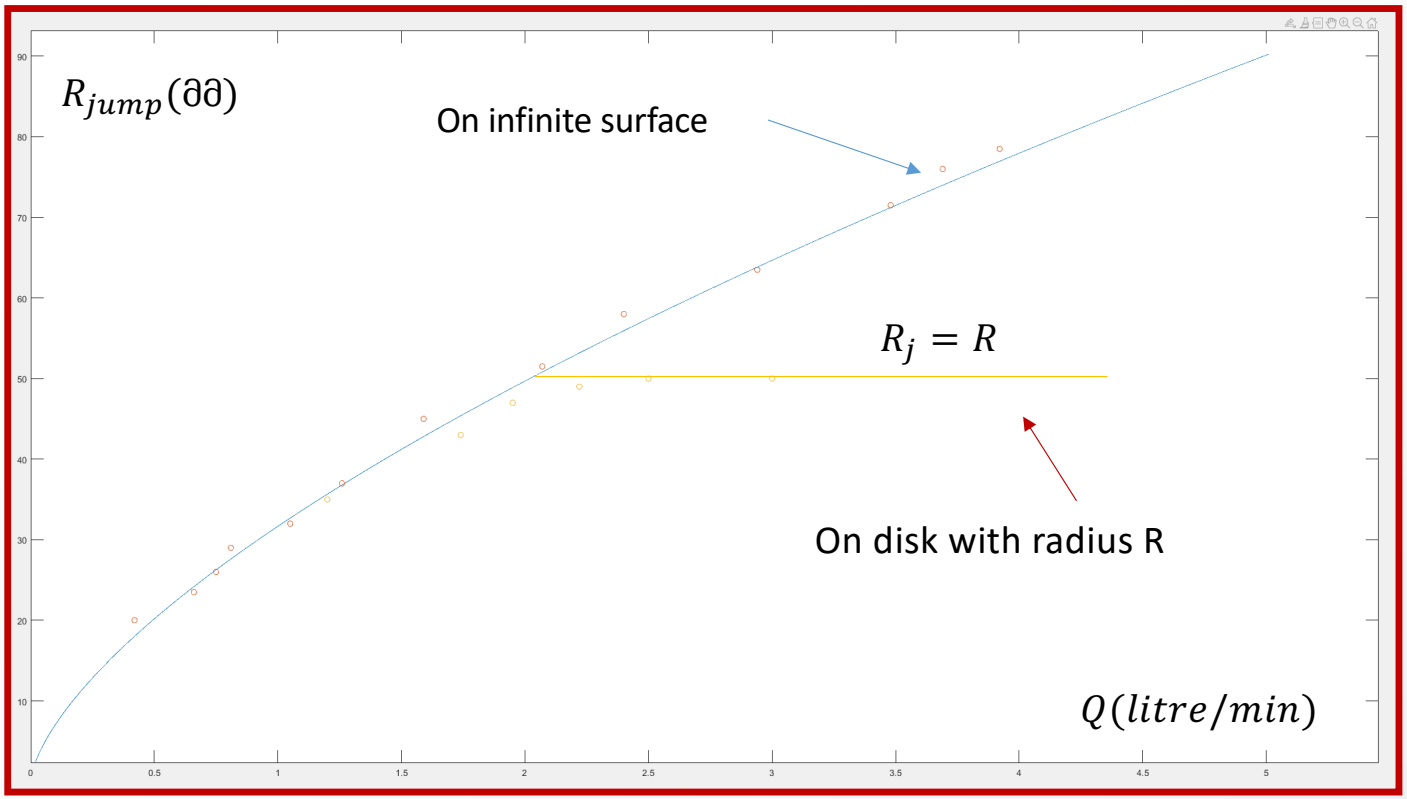


Sink swithed on

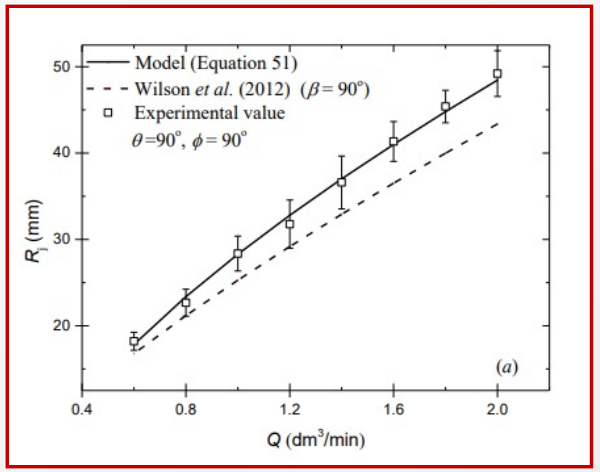
Depth reduced slowly to 1 cm

$$average\ force\colon \Delta F = 0.02\text{ნ} \ll mg$$

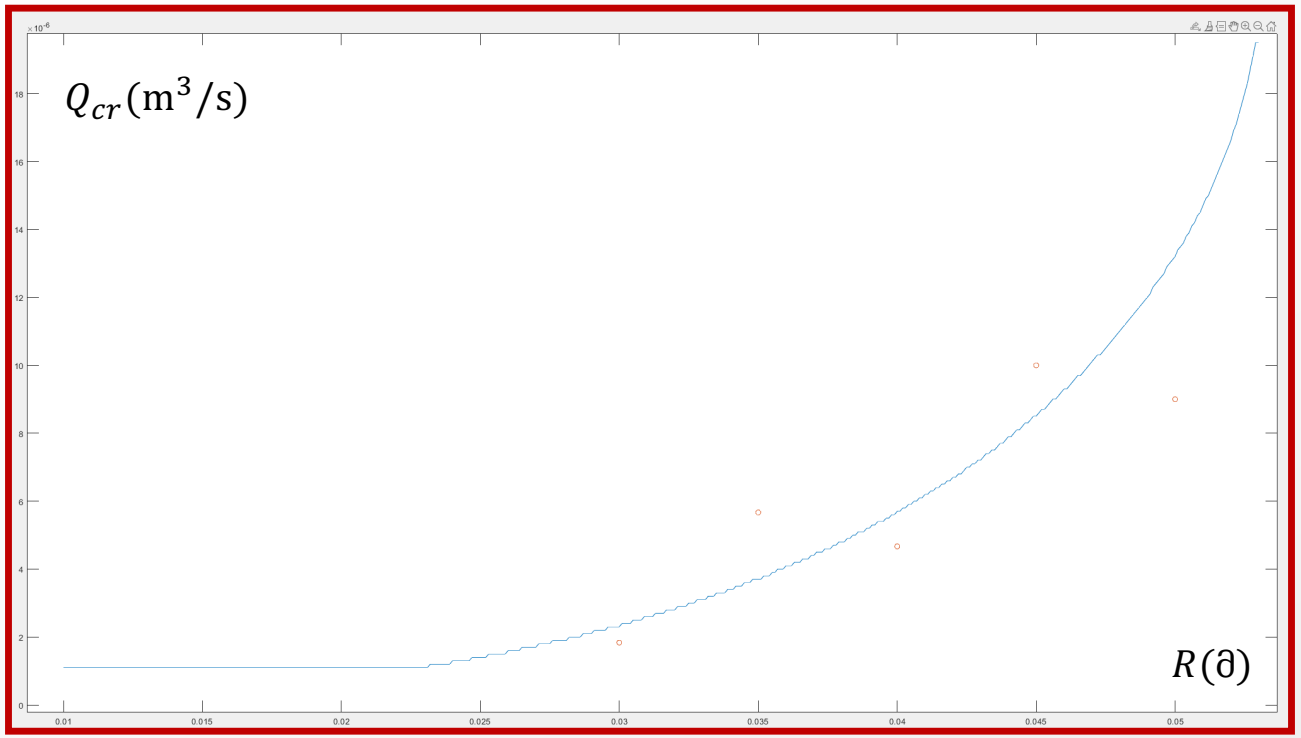
R_j vs Q



R.K. Bhagat and D.I. Wilson(2016)



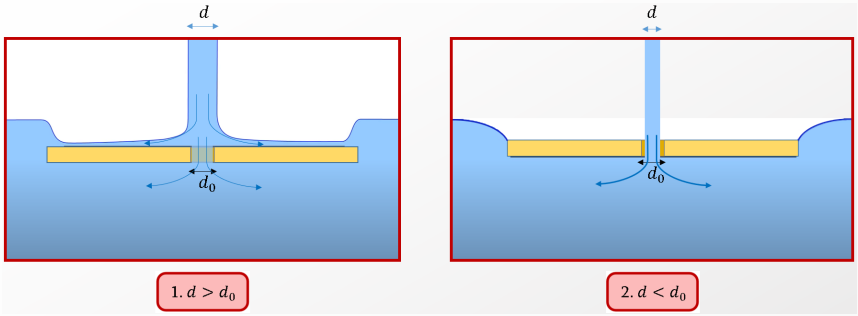
კრიტიკული ნაკადის რადიუსზე დამოკიდებულება



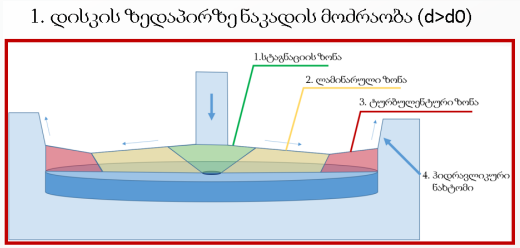
$$H_{j_0} = 1mm - 2.5mm$$

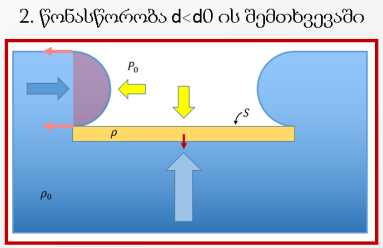
Conclusion

1) Explained phenomenon in 2 cases



2) *Developed theoretical model*



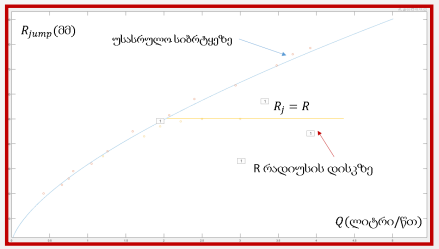


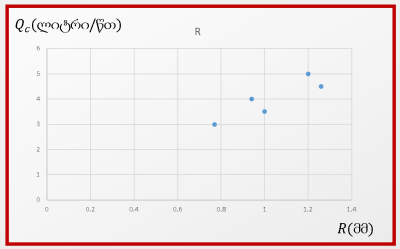
3) Developed Experimental setup and data taking





4) Compared experimental and theoretical results





Thanks for attention



```
clc;
clear;

d = 0.003;      %წაკადის დიამეტრი
r0 = 0;     %დისკის შიდა რადიუსი
d0 = 2*r0;
%R0 = 1;     %დისკის გარე რადიუსი

q0 = 1000;
q = 1250;
h = 0.001;

v = 1*10^(-3);
sigma = 0.07275;
g = 9.81;
P0 = 101325;

Q(1) = 0.000005;

Rej(1)=4*q0/pi/v/d*(1-(d0/d)^2)*Q(1);


rt(1) = 0.2964*d*Rej(1).^(1/3);


dr = 0.0001;
```

```

for(i = 1:500)


    r = rt(i);
    M = 32/79*q0*Q(i).^2/pi^2/d^2/r/(0.167/Rej(i).^0.25*(r/d)^2.25 + 2.37 - 0.0108*Rej(i).^0.5);

    while(M > sigma)
        M = 32/79*q0*Q(i).^2/pi^2/d^2/r/(0.167/Rej(i).^0.25*(r/d)^2.25 + 2.37 - 0.0108*Rej(i).^0.5);
        r = r + dr;
    end
    Rj(i) = r;

    Q(i+1)=Q(i)+0.000002;
    Rej(i+1) = 4*q0/pi/v/d*(1-(d0/d)^2)*Q(1+1);
    rt(i+1) = 0.2964*d*Rej(i+1).^(1/3);
end

Rj(501) = Rj(500);
plot(Q, Rj);

XE=xlsread('C:\Users\Home\Desktop\unsinkable_disk.xlsx','','D2:D15');
YE=xlsread('C:\Users\Home\Desktop\unsinkable_disk.xlsx','','E2:E15');

hold on;
scatter(XE, YE);
```



```
18
19 -    R(1) = R0;
20
21
22 -    N = 100;
23 -    dR = 0.001;
24 -    dr = 0.00001;
25
26 -  for (i = 1:N)
27
28 -      Q=0.000001;
29 -      Rej = 4*q0/pi/v/d*(1-(d0/d)^2)*Q;
30
31 -      Hj = 0;
32 -      H = Hj0;
33
34 -      while(Hj < H)
35
36 -          r = 0.2964*d*Rej^(1/3);
37
38 -          M = 32/79*q0*Q^2/pi^2/d^2/r/(0.167/Rej^0.25*(r/d)^2.25 + 2.37 - 0.0108*Rej^0.5);
39
40
41 -          while(r < R(i)  && M > sigma)
42
43 -              r = r + dr;
44
45 -              M = 32/79*q0*Q^2/pi^2/d^2/r/(0.167/Rej^0.25*(r/d)^2.25 + 2.37 - 0.0108*Rej^0.5);
46 -          end
47
48 -          if(M <= sigma)
49 -              Rj = r;
50 -              H = (q/q0-1)*R(i)^2/Rj^2*h + 4*Q^2/(pi^2*(d^2-d0^2)*g*Rj^2);
51 -              Hj = Hj0;
52
53 -          else if(r >= R(i))
54 -                  H = (q/q0-1)*h + 4*Q^2/(pi^2*(d^2-d0^2)*g*R(i)^2);
55 -                  Hj = ((M/sigma)-1)*R(i);
56 -              end
57 -          end
58 -          Q = Q + 0.000001;
59 -          Rej = 4*q0/pi/v/d*(1-(d0/d)^2)*Q;
60
61 -      end
62 -      Qcr(i) = Q;
63 -      R(i+1)=R(i)+dR;
64 -  end
65
66 -  Qcr(N+1) = Qcr(N);
67
68 -  plot(R, Qcr);
```

